

Intelligent Learning Assistant with LangGraph Agents

Overview

The Teaching assistant project is built on top of langraph agents and we have used multi tool implementation for different user prompt and we have used single llm to give the response based on the tool format we have uploaded in the code.

Implementation workflow

- We have used groq Api for model building and here we used LLama instant model.
- we have implemented the dynamic user filed inputs datas
- Instead of using large vertors here we will split the dataset into chunks.
- The chunck Splitting will be done for each response by doing that we can get accurate response.

- We have used PDF reader library to get the data from user and further processing units.
- We have used 5 tool to integrate with the multi definitions on each user response.
- we have used different formats here so that we can get response output differently for QUIZ,Assignment and summary.

INSTALLATION LIBRARY:

- langgraph
- langchain_community
- langchain_core
- sentence_transformers
- pypdf
- Pandas,OS,Fitz,,Streamlit,pptx